```
!pip install -q gradio

import gradio as gr
import json
import re
from datetime import datetime
from typing import Tuple, Dict, List
import random
```

```
from transformers import AutoTokenizer, AutoModelForCausalLM

tokenizer = AutoTokenizer.from_pretrained("ibm-granite/granite-3.3-2b-instruct")
model = AutoModelForCausalLM.from_pretrained("ibm-granite/granite-3.3-2b-instruct")
messages = [
    {"role": "user", "content": "Who are you?"},
]
inputs = tokenizer.apply_chat_template(
    messages,
    add_generation_prompt=True,
    tokenize=True,
    return_dict=True,
    return_tensors="pt",
).to(model.device)

outputs = model.generate(**inputs, max_new_tokens=40)
print(tokenizer.decode(outputs[0][inputs["input_ids"].shape[-1]:]))

!pip install -q gradio

import gradio as gr
import json
import re
from datetime import datetime
from typing import Tuple, Dict, List
import random
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
```

config.json: 100% 787/787 [00:00<00:00, 51.0kB/s]

tokenizer_config.json: 9.93k/? [00:00<00:00, 553kB/s]

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster d
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to en
```

vocab.json: 777k/? [00:00<00:00, 24.6MB/s]

merges.txt: 442k/? [00:00<00:00, 19.3MB/s]

added_tokens.json: 100% 207/207 [00:00<00:00, 17.0kB/s]

special_tokens_map.json: 100% 801/801 [00:00<00:00, 56.7kB/s]

tokenizer.json: 3.48M/? [00:00<00:00, 65.0MB/s]

model.safetensors.index.json: 29.8k/? [00:00<00:00, 1.54MB/s]

Download complete: 100% 5.07G/5.07G [01:14<00:00, 97.0MB/s]

Fetching 2 files: 100% 2/2 [01:04<00:00, 64.21s/it]

Loading weights: 100% 362/362 [00:02<00:00, 284.14it/s, Materializing param=model.norm.weight]

generation_config.json: 100% 132/132 [00:00<00:00, 5.36kB/s]

```
I am an assistant designed to help answer your questions and provide information. I strive to provide concise and accurate respo
```

```
class ColorPaletteGenerator:
    """Generate visually cohesive color palettes"""

    @staticmethod
    def generate_palette(style: str = "modern") -> Dict[str, str]:
```

```
    """Generate a cohesive color palette based on style"""

    palettes = {
        'modern': {
            'primary': '#667eea',
            'secondary': '#764ba2',
            'accent': '#f093fb',
            'background': '#0f172a',
            'surface': '#1e293b',
            'text_primary': '#f1f5f9',
            'text_secondary': '#cbd5e1',
            'border': '#334155'
        },
        'vibrant': {
            'primary': '#ff6b6b',
            'secondary': '#ff8e72',
            'accent': '#f5576c',
            'background': '#0f172a',
            'surface': '#1e293b',
            'text_primary': '#f1f5f9',
            'text_secondary': '#cbd5e1',
            'border': '#334155'
        },
        'ocean': {
            'primary': '#0099cc',
            'secondary': '#006699',
            'accent': '#00ccff',
            'background': '#0f172a',
            'surface': '#1e293b',
            'text_primary': '#f1f5f9',
            'text_secondary': '#cbd5e1',
            'border': '#334155'
        },
        'forest': {
            'primary': '#27ae60',
            'secondary': '#1e8449',
            'accent': '#52be80',
            'background': '#0f172a',
            'surface': '#1e293b',
            'text_primary': '#f1f5f9',
            'text_secondary': '#cbd5e1',
            'border': '#334155'
        },
        'sunset': {
            'primary': '#ff8c42',
            'secondary': '#ff6b35',
            'accent': '#ffa751',
            'background': '#0f172a',
            'surface': '#1e293b',
            'text_primary': '#f1f5f9',
            'text_secondary': '#cbd5e1',
            'border': '#334155'
        },
        'luxury': {
            'primary': '#1a1a1a',
            'secondary': '#333333',
            'accent': '#FFD700',
            'background': '#0f172a',
            'surface': '#1e293b',
            'text_primary': '#f1f5f9',
            'text_secondary': '#cbd5e1',
            'border': '#334155'
        },
        'creative': {
            'primary': '#9b59b6',
            'secondary': '#8e44ad',
            'accent': '#e74c3c',
            'background': '#0f172a',
            'surface': '#1e293b',
            'text_primary': '#f1f5f9',
            'text_secondary': '#cbd5e1',
            'border': '#334155'
        },
        'minimal': {
            'primary': '#2c3e50',
            'secondary': '#34495e',
            'accent': '#3498db',
```

```
                'background': '#0f172a',
                'surface': '#1e293b',
                'text_primary': '#f1f5f9',
                'text_secondary': '#cbd5e1',
                'border': '#334155'
            }
        }

        return palettes.get(style.lower(), palettes['modern'])
```

```
class HTMLPageBuilder:
    """Build styled HTML pages from descriptions"""

    @staticmethod
    def extract_sections(description: str) -> List[str]:
        """Extract sections from description"""
        sections = []

        section_keywords = {
            'hero': ['hero', 'banner', 'landing'],
            'features': ['features', 'services', 'benefits'],
            'about': ['about', 'team', 'company'],
            'portfolio': ['portfolio', 'gallery', 'work', 'projects'],
            'testimonials': ['testimonial', 'reviews', 'feedback'],
            'pricing': ['pricing', 'plans', 'cost'],
            'contact': ['contact', 'form', 'reach'],
            'footer': ['footer', 'bottom']
        }

        desc_lower = description.lower()
```

```
from typing import Dict, List

class HTMLGenerator:

    @staticmethod
    def detect_sections(description: str) -> List[str]:
        """Detect sections from description"""

        section_keywords = {
            "hero": ["welcome", "hero", "intro"],
            "features": ["features", "services", "benefits"],
            "about": ["about", "company", "who we are"],
            "portfolio": ["portfolio", "work", "projects"],
            "testimonials": ["testimonials", "reviews", "clients"],
            "pricing": ["pricing", "plans", "cost"],
            "contact": ["contact", "email", "phone"]
        }

        desc_lower = description.lower()
        sections = []

        for section_type, keywords in section_keywords.items():
            for keyword in keywords:
                if keyword in desc_lower:
                    sections.append(section_type)
                    break

        if not sections:
            sections = ['hero', 'features', 'contact', 'footer']

        return sections

    @staticmethod
    def generate_meta_tags(title: str, description: str) -> str:
        """Generate SEO meta tags"""

        meta_tags = f"""
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>{title}</title>
    <meta name="description" content="{description[:160]}">
    <meta name="theme-color" content="#0f172a">
    <meta property="og:title" content="{title}">
    <meta property="og:description" content="{description[:160]}">
    <meta property="og:type" content="website">
```

```
        <meta name="twitter:card" content="summary_large_image">
        <meta name="author" content="HTML Quick-Styler">
    """
            return meta_tags

        @staticmethod
        def generate_section_html(section_type: str, palette: Dict) -> str:
            """Generate HTML for specific section"""

            sections = {

                "hero": """
            <section class="hero">
                <div class="hero-content">
                    <h1>Welcome to Your Site</h1>
                    <p>Create amazing web experiences</p>
                    <button class="btn btn-primary">Get Started</button>
                </div>
            </section>
    """,

                "features": """
            <section class="features">
                <h2>Features</h2>
                <div class="features-grid">
                    <div class="feature-card">
                        <h3>Fast</h3>
                        <p>Lightning-quick performance</p>
                    </div>
                    <div class="feature-card">
                        <h3>Secure</h3>
                        <p>Enterprise-grade security</p>
                    </div>
                    <div class="feature-card">
                        <h3>Scalable</h3>
                        <p>Grow without limits</p>
                    </div>
                </div>
            </section>
    """,

                "about": """
            <section class="about">
                <div class="about-content">
                    <h2>About Us</h2>
                    <p>We're passionate about creating beautiful, functional websites that help your business grow.</p>
                </div>
            </section>
    """,

                "contact": """
            <section class="contact">
                <h2>Get in Touch</h2>
                <form class="contact-form">
                    <input type="text" placeholder="Your Name" required>
                    <input type="email" placeholder="Your Email" required>
                    <textarea placeholder="Your Message" rows="5" required></textarea>
                    <button type="submit" class="btn btn-primary">Send Message</button>
                </form>
            </section>
    """,

                "footer": """
            <footer class="footer">
                <div class="footer-content">
                    <p>&copy; 2024 Your Company. All rights reserved.</p>
                    <div class="footer-links">
                        <a href="#">Privacy</a>
                        <a href="#">Terms</a>
                        <a href="#">Contact</a>
                    </div>
                </div>
            </footer>
    """
            }

            return sections.get(section_type, "")
```

```
    @staticmethod
    def generate_css(palette: Dict) -> str:
        """Generate CSS"""

        css = f"""
<style>

:root {{
    --primary: {palette.get('primary', '#3b82f6')};
    --secondary: {palette.get('secondary', '#6366f1')};
    --accent: {palette.get('accent', '#22c55e')};
    --background: {palette.get('background', '#0f172a')};
    --surface: {palette.get('surface', '#1e293b')};
    --text-primary: {palette.get('text_primary', '#ffffff')};
    --text-secondary: {palette.get('text_secondary', '#cbd5f5')};
    --border: {palette.get('border', '#334155')};
}}

body {{
    font-family: Arial, sans-serif;
    background: var(--background);
    color: var(--text-primary);
}}

.hero {{
    padding: 80px;
    text-align: center;
    background: linear-gradient(135deg, var(--primary), var(--secondary));
}}

.features-grid {{
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(250px,1fr));
    gap: 20px;
}}

.feature-card {{
    background: var(--surface);
    padding: 20px;
    border-radius: 10px;
}}

.footer {{
    text-align: center;
    padding: 30px;
    background: var(--surface);
}}

</style>
"""

    @staticmethod
    def generate_css(palette):
        css = f"""
<style>
body {{
    font-family: Arial, sans-serif;
    margin:0;
    padding:0;
    background:{palette['background']};
}}

header {{
    background:{palette['primary']};
    color:white;
    padding:20px;
}}

nav {{
    display:flex;
    justify-content:space-between;
    align-items:center;
}}

nav a {{
    color:white;
```

```
        margin-left:15px;
        text-decoration:none;
    }}

    section {{
        padding:40px;
    }}

    .features {{
        background:{palette['secondary']};
    }}

    .contact {{
        background:{palette['accent']};
    }}

    footer {{
        text-align:center;
        padding:20px;
    }}
    </style>
    """
            return css
```

```
class HTMLGenerationEngine:
    """Main engine for HTML page generation"""

    @staticmethod
    def generate_page(title: str, description: str, style: str = "modern") -> Tuple[str, str]:
        """Generate complete HTML page"""

        # Generate color palette
        palette = ColorPaletteGenerator.generate_palette(style)

        # Extract sections
        sections = HTMLPageBuilder.extract_sections(description)

        # Generate meta tags
        meta_tags = HTMLPageBuilder.generate_meta_tags(title, description)

        # Generate sections
        sections_html = '\n\n'.join([
            HTMLPageBuilder.generate_section_html(section, palette)
            for section in sections
            if HTMLPageBuilder.generate_section_html(section, palette)
        ])
```

```
    css = HTMLPageBuilder.generate_css(palette)

    # Build complete HTML
    html = f'''<!DOCTYPE html>
    <html lang="en">
    <head>
    {meta_tags}
    {css}
    </head>
    <body>
        <header>
            <nav>
                <h1>{title}</h1>
                <div>
                    <a href="#features">Features</a>
                    <a href="#about">About</a>
                    <a href="#contact">Contact</a>
                </div>
            </nav>
        </header>

        <main>
    {sections_html}
        </main>
    </body>
    </html>'''

    return html, palette
```

```
generation_history = []


def generate_html_page(title: str, description: str, style: str = "modern") -> Tuple[str, str, str]:
    """Generate HTML page with styling"""

    if not title.strip() or not description.strip():
        return "", "❌ Please provide both title and description!", ""

    try:
        html_output, palette = HTMLGenerationEngine.generate_page(title, description, style)

        color_info = f"""
**Color Palette:**
- Primary: {palette['primary']}
- Secondary: {palette['secondary']}
- Accent: {palette['accent']}
- Background: {palette['background']}
"""

        generation_history.append({
            'title': title,
            'description': description,
            'html': html_output,
            'palette': palette,
            'timestamp': datetime.now().isoformat()
        })

        return html_output, "✅ HTML page generated successfully!", color_info

    except Exception as e:
        return "", f"❌ Error: {str(e)}", ""
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
/tmp/ipykernel_238/2411705232.py in <cell line: 0>()
----> 1 css = HTMLPageBuilder.generate_css(palette)
      2
      3 # Build complete HTML
      4 html = f'''<!DOCTYPE html>
      5 <html lang="en">

AttributeError: type object 'HTMLPageBuilder' has no attribute 'generate_css'
```

Next steps: Explain error

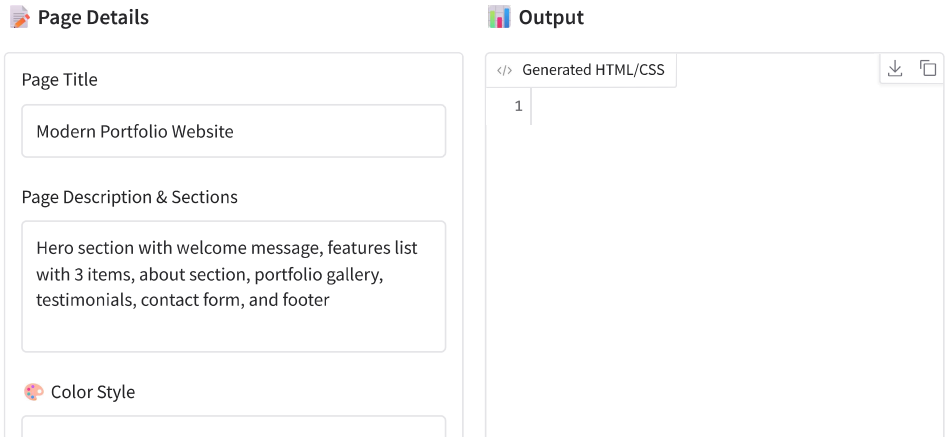
```
def get_color_suggestions(style: str) -> str:
    """Get color suggestions based on style"""
    palette = ColorPaletteGenerator.generate_palette(style)

    palette_text = f"""
**Palette for "{style}" style:**
- Primary: {palette['primary']}
- Secondary: {palette['secondary']}
- Accent: {palette['accent']}
- Background: {palette['background']}
- Text: {palette['text_primary']}
"""
    return palette_text

print("🚀 Loading HTML Quick-Styler Advanced...")

with gr.Blocks(title="HTML Quick-Styler - Advanced Edition") as demo:

    gr.Markdown("""
    # 🎨 HTML Quick-Styler Advanced
    ## Professional HTML Page Generator with Auto-Styling

    **Generate production-ready HTML pages in seconds**
    """)

    with gr.Tabs():

        # ============= TAB 1: Page Generator =============
        with gr.TabItem("🎯 Generate Page"):
```

```
        with gr.Row():
            with gr.Column(scale=1):
                gr.Markdown("### 📝 Page Details")

                page_title = gr.Textbox(
                    label="Page Title",
                    value="Modern Portfolio Website",
                    lines=1
                )

                page_description = gr.Textbox(
                    label="Page Description & Sections",
                    value="Hero section with welcome message, features list with 3 items, about section, portfolio gallery,
                    lines=4
                )

                color_style = gr.Dropdown(
                    ["modern", "vibrant", "ocean", "forest", "sunset", "luxury", "creative", "minimal"],
                    value="modern",
                    label="🎨 Color Style"
                )

                generate_btn = gr.Button("✨ Generate HTML", size="lg", variant="primary")

            with gr.Column(scale=1):
                gr.Markdown("### 📊 Output")

                html_output = gr.Code(
                    language="html",
                    label="Generated HTML/CSS",
                    lines=20
                )

                status_msg = gr.Markdown()
                color_info = gr.Markdown()

        generate_btn.click(
            generate_html_page,
            inputs=[page_title, page_description, color_style],
            outputs=[html_output, status_msg, color_info]
        )

    # ============= TAB 2: Preview =============
    with gr.TabItem("👁️ Live Preview"):
        gr.Markdown("### 🔍 Preview Generated Page")

        preview_button = gr.Button("🔄 Load Preview", size="lg", variant="primary")
        preview_output = gr.HTML(value="<p style='text-align:center; color:#fff; padding:40px;'>No preview loaded. Generate

        def show_preview(html_text):
            if not html_text.strip():
                return "<p style='text-align:center; color:#fff; padding:40px;'>No HTML provided</p>"
            return html_text

        preview_button.click(
            show_preview,
            inputs=[html_output],
            outputs=[preview_output]
        )

    # ============= TAB 3: Color Palettes =============
    with gr.TabItem("🎨 Color Palettes"):
        gr.Markdown("### 🌈 Explore Color Schemes")

        with gr.Row():
            style_selector = gr.Radio(
                ["modern", "vibrant", "ocean", "forest", "sunset", "luxury", "creative", "minimal"],
                value="modern",
                label="Choose Style"
            )

            palette_btn = gr.Button("🎨 Get Palette", size="lg", variant="primary")

        palette_display = gr.Markdown()

        palette_btn.click(
            get_color_suggestions,
```

```
            inputs=[style_selector],
            outputs=[palette_display]
        )

    # ============= TAB 4: Templates =============
    with gr.TabItem("📋 Templates"):
        gr.Markdown("### 📝 Quick Templates")

        with gr.Row():
            with gr.Column():
                gr.Markdown("""
                #### Portfolio Website
                Hero, Portfolio Gallery, About, Contact, Footer
                """)
                portfolio_btn = gr.Button("Use Template", size="sm")

                def load_portfolio():
                    return ("Portfolio Website",
                            "Hero section with welcome banner, portfolio gallery with 6 items, about section, testimonials,
                            "modern")

                portfolio_btn.click(load_portfolio, outputs=[page_title, page_description, color_style])

            with gr.Column():
                gr.Markdown("""
                #### SaaS Landing Page
                Hero, Features, Pricing, Testimonials, CTA, Footer
                """)
                saas_btn = gr.Button("Use Template", size="sm")

                def load_saas():
                    return ("SaaS Product",
                            "Hero banner, features showcase with 3 items, pricing plans comparison, testimonials, call-to-ac
                            "vibrant")

                saas_btn.click(load_saas, outputs=[page_title, page_description, color_style])

            with gr.Column():
                gr.Markdown("""
                #### Company Website
                Header, About, Services, Team, Contact, Footer
                """)
                company_btn = gr.Button("Use Template", size="sm")

                def load_company():
                    return ("Company",
                            "Header with navigation, hero section, about company, services list, team members, contact form,
                            "ocean")

                company_btn.click(load_company, outputs=[page_title, page_description, color_style])

    # ============= TAB 5: Guide =============
    with gr.TabItem("📚 Guide"):
        gr.Markdown("""
        # 📖 HTML Quick-Styler Guide

        ## How It Works

        ### 1. Enter Details
        - **Title:** Page name
        - **Description:** Sections you want
        - **Color Style:** Pick a theme

        ### 2. Automatic Detection
        The tool recognizes these keywords:
        - `hero` / `banner` → Hero Section
        - `features` / `services` → Features Grid
        - `about` / `team` → About Section
        - `portfolio` / `gallery` → Portfolio
        - `testimonial` / `reviews` → Testimonials
        - `pricing` / `plans` → Pricing Plans
        - `contact` / `form` → Contact Form
        - `footer` → Footer

        ### 3. Generate & Customize
        - Copy generated HTML
        - Edit text and colors
```

```
            - Save as .html file
            - Open in browser

            ## Features
            ✅ Responsive design
            ✅ Modern animations
            ✅ SEO meta tags
            ✅ Professional styling
            ✅ Dark theme
            ✅ Color palettes
            ✅ Copy-paste ready

            ## Color Styles
            - **Modern:** Professional blues
            - **Vibrant:** Bold reds
            - **Ocean:** Calm blues
            - **Forest:** Natural greens
            - **Sunset:** Warm oranges
            - **Luxury:** Elegant blacks
            - **Creative:** Purple artistic
            - **Minimal:** Clean grays
            """)

print("\n✅ HTML Quick-Styler Advanced Ready!\n")
demo.launch(share=True, show_error=True, show_api=False)
```

```
🚀 Loading HTML Quick-Styler Advanced...
/tmp/ipykernel_238/4188347734.py:208: DeprecationWarning: The 'show_api' parameter in launch() will be removed in Gradio 6.0. Yo
  demo.launch(share=True, show_error=True, show_api=False)

✅ HTML Quick-Styler Advanced Ready!

Colab notebook detected. To show errors in colab notebook, set debug=True in launch()
* Running on public URL: https://8c77af6a885eed97fc.gradio.live

This share link expires in 1 week. For free permanent hosting and GPU upgrades, run `gradio deploy` from the terminal in the wor
```



Next steps: Deploy to Cloud Run